

CO2 ASSESSMENT TOOL- DEBUGGING

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COURSE CODE – CSA0307

COURSE NAME – DATA STRUCTURES

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Q3
8. Buggy code in node creation
10 marks
3/3 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int linearSearch(int arr[], int n, int key) {
3     for(int i=0; i<n; i++)
4     {
5         if(arr[i]==key)
6             return i;
7     }
8     return -1;
9 }
10 int main()
11 {
12     int arr[]={10,20,30,40,50};
13     int key=30;
14     int result=linearSearch(arr,5,key);
15     if(result!=-1)
16         printf("element not found");
17     else
18         printf("element found at index %d", result);
19     return 0;
20 }
21 }
22 }
```

Run Save Submit Fix

Test Input
stdin input (optional)

Output
element found at index 2

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QUESTIONS
Q1
1. Find the bugs; include the missing lines in the following snippet.
10 marks
Q2
2. Buggy Code in an array
10 marks
Q3
8. Buggy code in node creation
10 marks
3/3 answered

Q2 2. Buggy Code in an array 10 marks

```
#include <stdio.h>
int main() {
    int arr[5] = {10,20,30,40,50};
    for(int i=0; i<=5; i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Expected output:
10 20 30 40 50

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int arr[5]={10,20,30,40,50};
5     for(int i=0; i<5; i++)
6     {
7         printf("%d", arr[i]);
8     }
9     return 0;
10 }
```

Test Input
stdin input (optional)

Output
1020304050

10 marks

3/3 answered

```
newNode->data = 10;  
newNode->next = NULL;  
  
printf("%d", newNode->data);  
return 0;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 #include<stdlib.h>  
3 struct Node  
4 {  
5     int data;  
6     struct Node *next;  
7 };  
8 int main()  
9 {  
10     struct Node *newNode;  
11     newNode=(struct Node  
12     *)malloc(sizeof(struct Node));  
13     newNode->data=10;  
14     newNode->next=NULL;  
15     printf("%d", newNode->data);  
16     free(newNode);  
17     return 0;  
18 }
```

Test Input

stdin input (optional)

Output

10



Run



Save



Submit Fix